

1 • Shared evidence: dispersal jury

The seed dispersal design lab

Name _____ Date _____

All fall times below are constructed sample data, not class observations.

A • Sycamore

A seed is enclosed in the base of a winged fruit. The wing can slow descent and help wind carry it.

C • Fleshy fruit

Some animals eat fruit, digest the flesh and later pass surviving seeds elsewhere. Survival is not guaranteed.

B • Burdock

Hooked burrs can attach to animal fur and be carried away before falling off.

D • Test method

Constructed test: same paper stock and original 4 × 16 cm size, all paper retained; same 1.5 m release height and rule, still indoor air. Shape changes. Fall time is measured.

Model	Trial 1 / s	Trial 2 / s	Trial 3 / s
Compact paper	0.8	0.9	0.7
Rotor paper	1.4	1.5	1.3

W1. Match A-C to a dispersal mechanism. W2. Compare one pair of times.

W3. Improve the “always travels farther” claim and name a limit.

2 • Supported route: structure and evidence

Use page 1. Point, circle or explain with a short sentence.

S1. A sycamore wing mainly helps interaction with ...

A. Air

B. An animal's digestive system

S2. Burdock hooks can help attachment to ...

A. Animal fur

B. A pollen tube

**S3. Complete: In trial 1, rotor paper took _____ s and compact paper took _____ s.
The _____ stayed in the air longer.**

S4. Name one condition kept the same in the paper test.

S5. Does this prove real outdoor travel distance? Explain.

3 • Standard route: design jury report

Use the shared evidence and state exactly what was measured.

N1. Explain a wing, hook and fleshy-fruit dispersal route.

N2. Compare the two models using two pairs of times and units.

N3. Name the changed variable, two controls and the measured quantity.

N4. Explain why adding a clip to one model weakens a shape-only comparison.

N5. Give a bounded conclusion and one model limitation.

4 • Stretch route: design a stronger test

Do not confuse fall time with distance.

T1. Calculate the mean fall time for each paper shape and compare them. Show working.

T2. Explain why the model supports a possible role for a wing but cannot prove every real seed travels farther.

T3. Design a later model test of horizontal distance using controlled airflow. Name the change, two controls, repeated measurement and units.

T4. Evaluate “plants grew hooks because they wanted animals to carry them”. Write a scientific replacement without intention.

5 • Exit and reflection

Use evidence and an honest limit.

E1. Explain one structure-dispersal link.

E2. Compare two fall times from page 1 using units.

E3. What does the paper model not establish?

E4. Name a control and explain its purpose.

Reflection. What should the next model measure?
